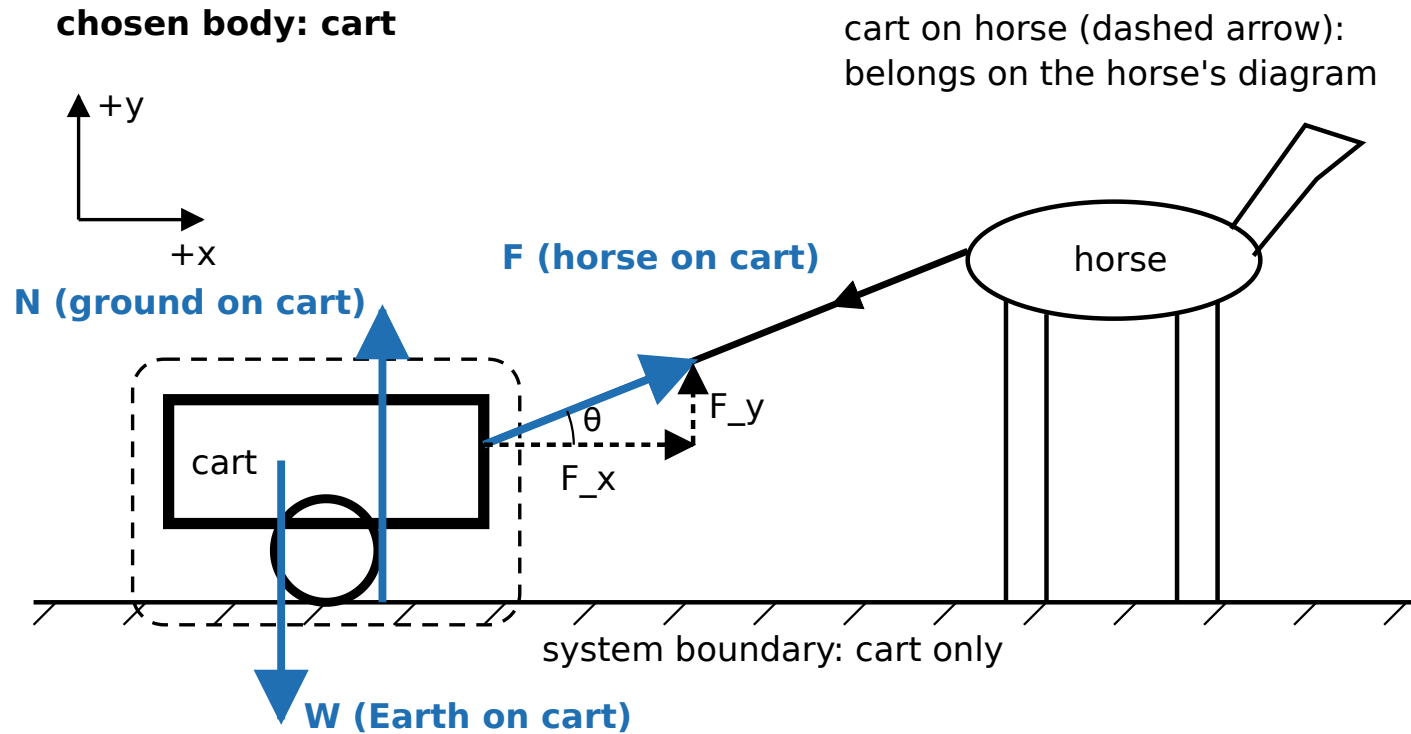


Authored Grade-9 Force and Laws of Motion spine covering balanced/unbalanced for: Reconstruction

Zero net force allows steady motion

Predict

A low-friction cart is already moving when the push ends. Must a new forward force keep acting just to preserve that motion?



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Attempt

A two-case prediction for a released moving cart and an initially resting cart under the same zero-net-force condition.

Reconstruct

1. After the hand leaves the cart, is the hand still exerting the forward push?
2. What would count as a change of velocity in the next few moments?
3. If none of those changes occurs, what part of the cart's motion is staying the same?
4. Does that same rule also allow a cart whose starting velocity is zero?

Diagnose

- A low-friction cart is pushed and then released. After the hand loses contact, what horizontal force is required merely to keep it moving steadily?

Repair

- Separate velocity from change of velocity: a net force is required for acceleration, not for the existence of motion.

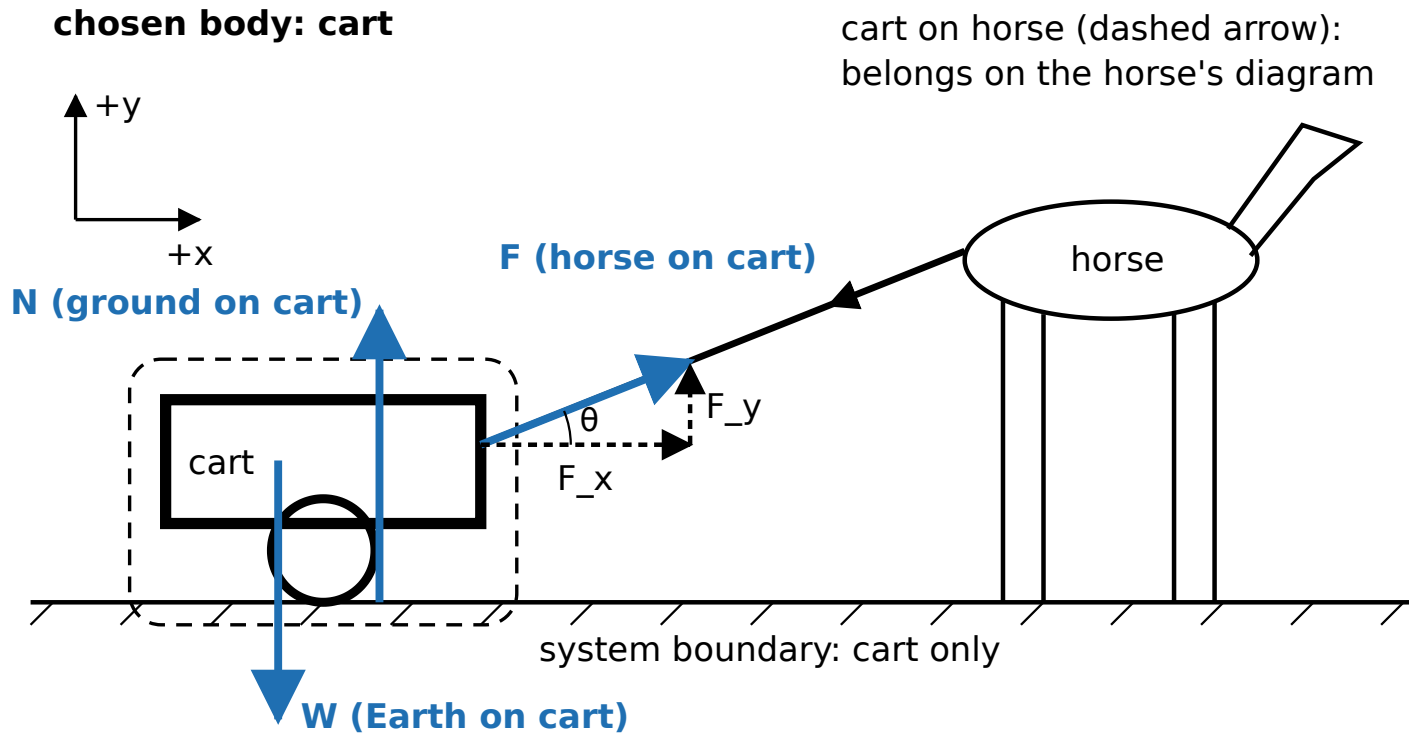
What a complete answer does

The moving cart is allowed to continue steadily.; The resting cart is allowed to remain at rest under the same rule.

- 'The moving cart keeps moving steadily; the resting cart stays at rest. Neither has to change velocity.'

The step you rebuilt

Zero net force does not mean zero motion; it means velocity does not change, so rest and steady motion are both compatible with the same condition.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Boundary test

A cart moving east begins to slow down. Can zero net force still describe that interval?

Boundary answer

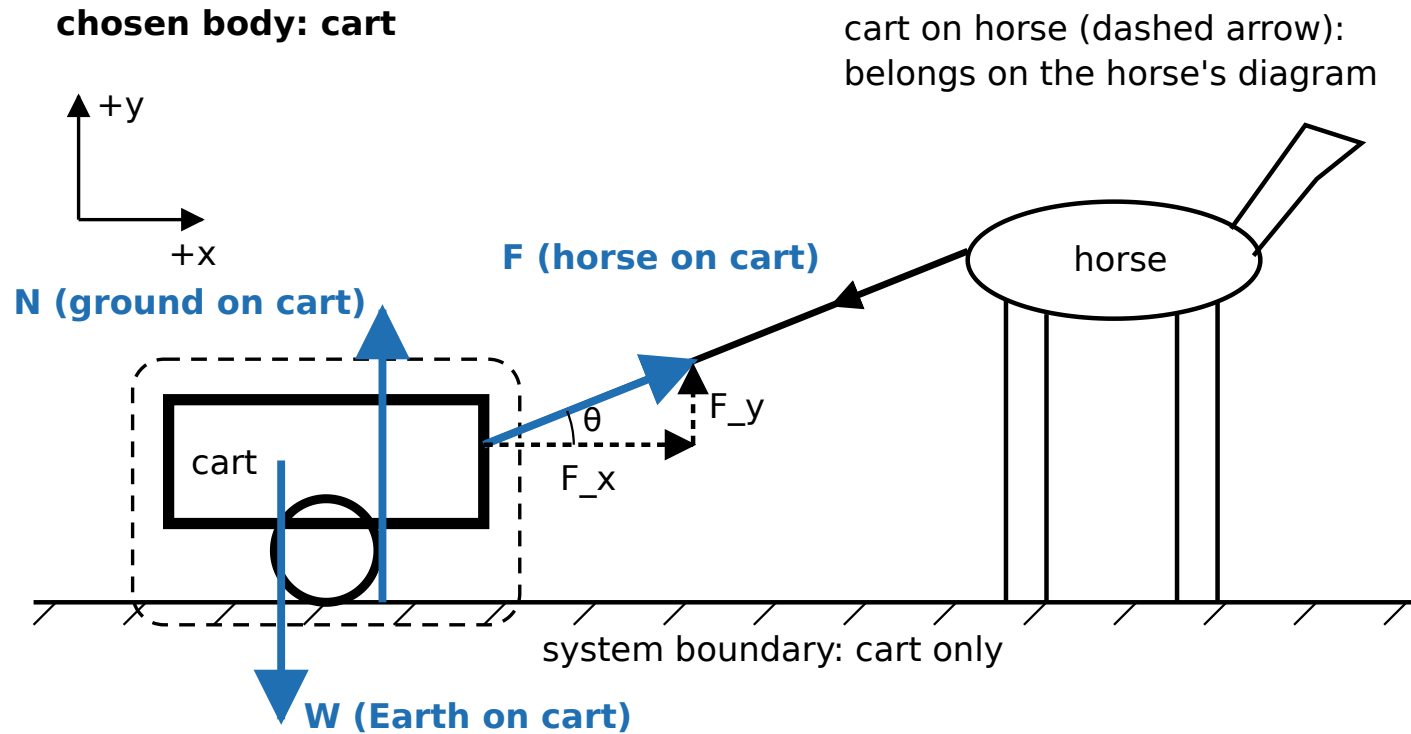
No. Slowing down is a change of velocity, so the zero-net-force condition no longer fits the stated interval.

The rule is about unchanged velocity, not simply about whether the object is moving.

Nonzero forces can cancel

Predict

Two people push the same box equally hard in opposite directions. If the box has zero net force, does that mean neither person is exerting a force?



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Attempt

A force inventory that keeps both opposite contributions visible plus a separate statement of the net result.

Reconstruct

1. Before adding anything, how many pushes are physically acting on the box?
2. Which directions do those pushes point?
3. If the two sizes match, what happens to the combined directional effect?
4. If both people push harder by the same amount, which statements change and which stay true?

Diagnose

- A book rests on a table. Are there no forces on the book, or can nonzero forces add to zero?

Repair

- List every interaction force on the body, then combine their directions; cancellation belongs to the net result, not to the individual forces.

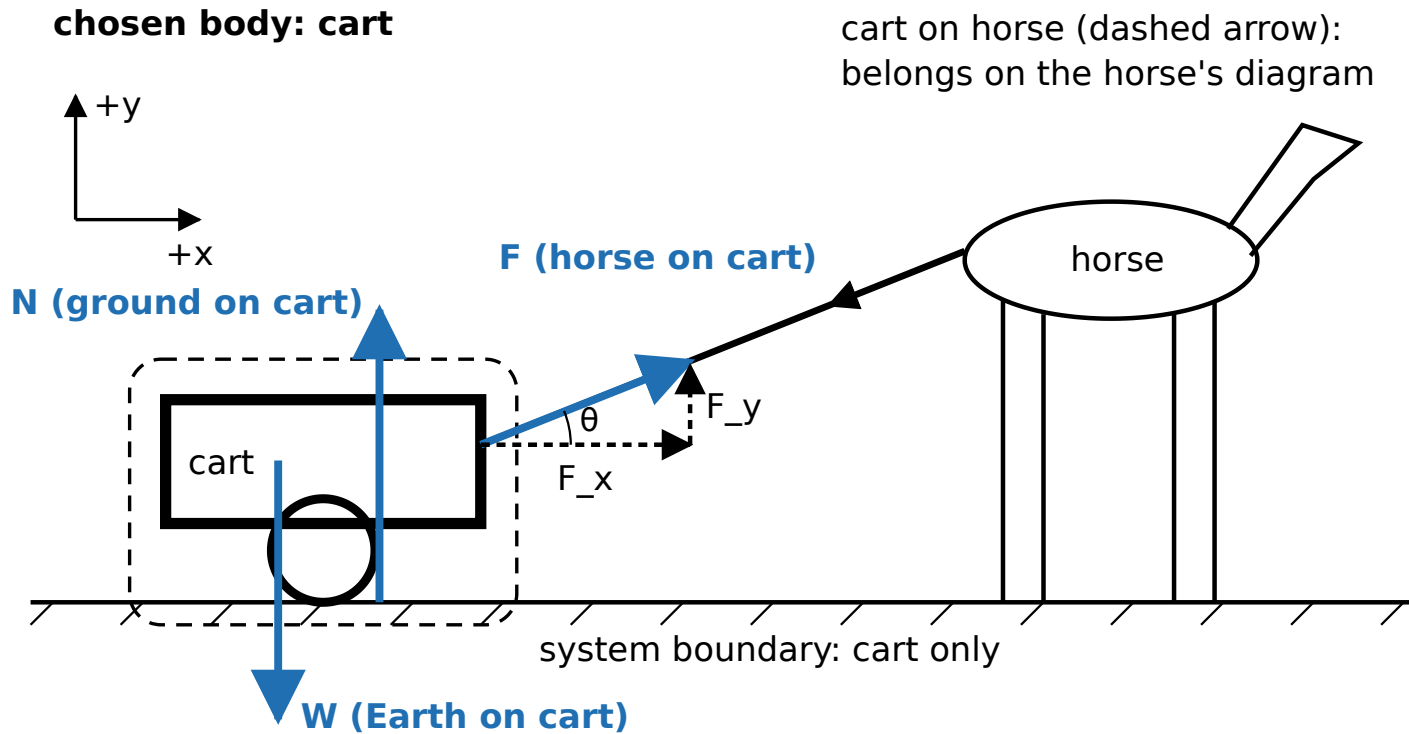
What a complete answer does

Both nonzero contributions remain listed.; Only the combined result is labelled zero.

- 'Left push present; right push present; equal and opposite, so the net force is zero.'

The step you rebuilt

Several nonzero forces can act on one body while their directed sum is zero; the motion responds to the net result rather than to the mere presence of forces.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Boundary test

One of the two equal opposite pushes is removed. Is the net force still zero?

Boundary answer

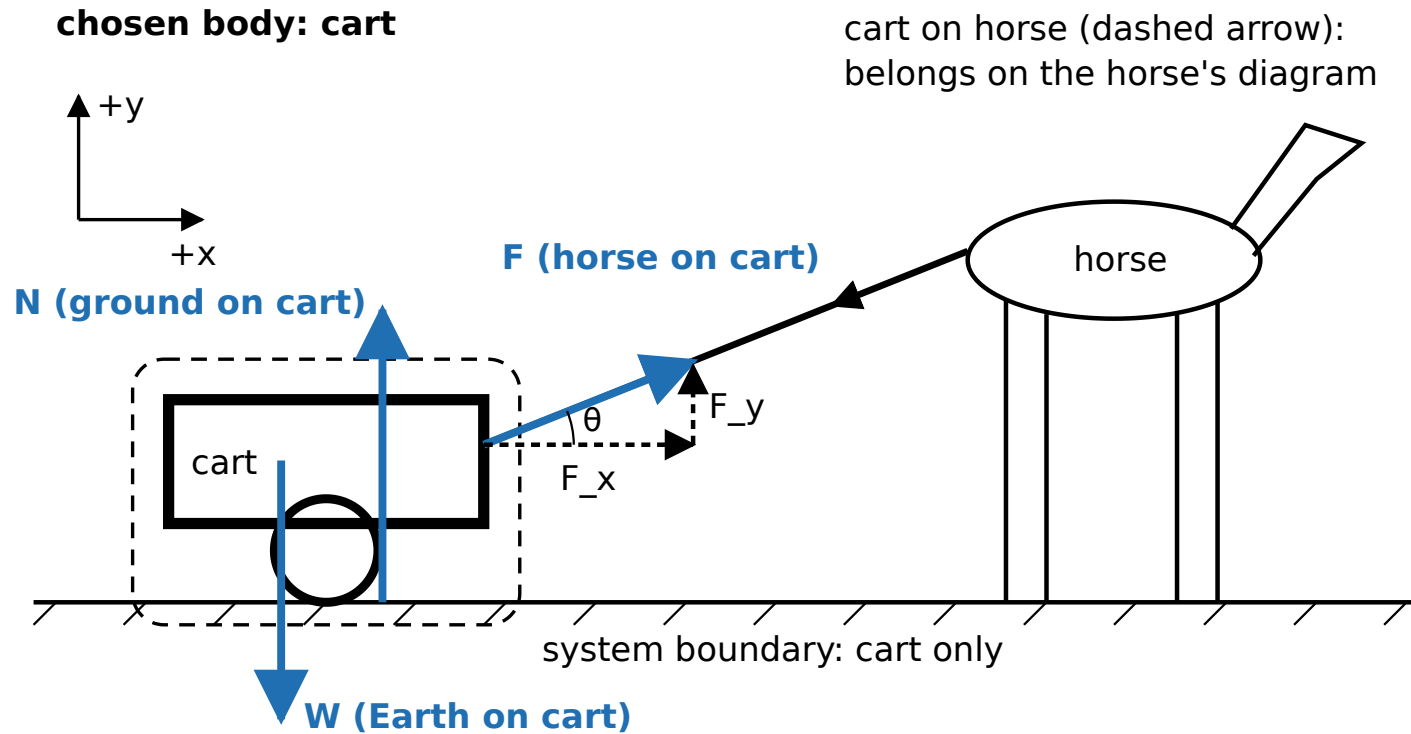
No. One nonzero force remains without an equal opposite contribution to cancel it.

Zero net force came from cancellation between present forces, not from a property of the object.

A free-body diagram belongs to one chosen body

Predict

A horse pulls a cart while the cart pulls back on the horse. If you draw only the cart, should both of those horizontal forces appear on the cart's free-body diagram?



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Attempt

Two separate body inventories, one for the cart and one for the horse, with the contact partner arrows assigned to the body each arrow acts on.

Reconstruct

1. Which single body are you describing first?
2. For the horse-on-cart force, which object receives the push?
3. For the cart-on-horse partner, which object receives that push?
4. When you switch the chosen body, which contact arrow moves into the new inventory?

Diagnose

- A horse pulls a cart and the cart pulls back equally. Which of those two forces acts on the cart, and which acts on the horse?
- If gravity is one Earth-on-block interaction, did choosing tilted axes create two new physical agents?

Repair

- Choose one body first. Put only forces acting on that body on its diagram; draw the partner force only when the other body is chosen.
- No. Weight remains one physical force. Its parallel/perpendicular components are the coordinate representation used in the chosen equations; do not add the original mg again to those component sums.

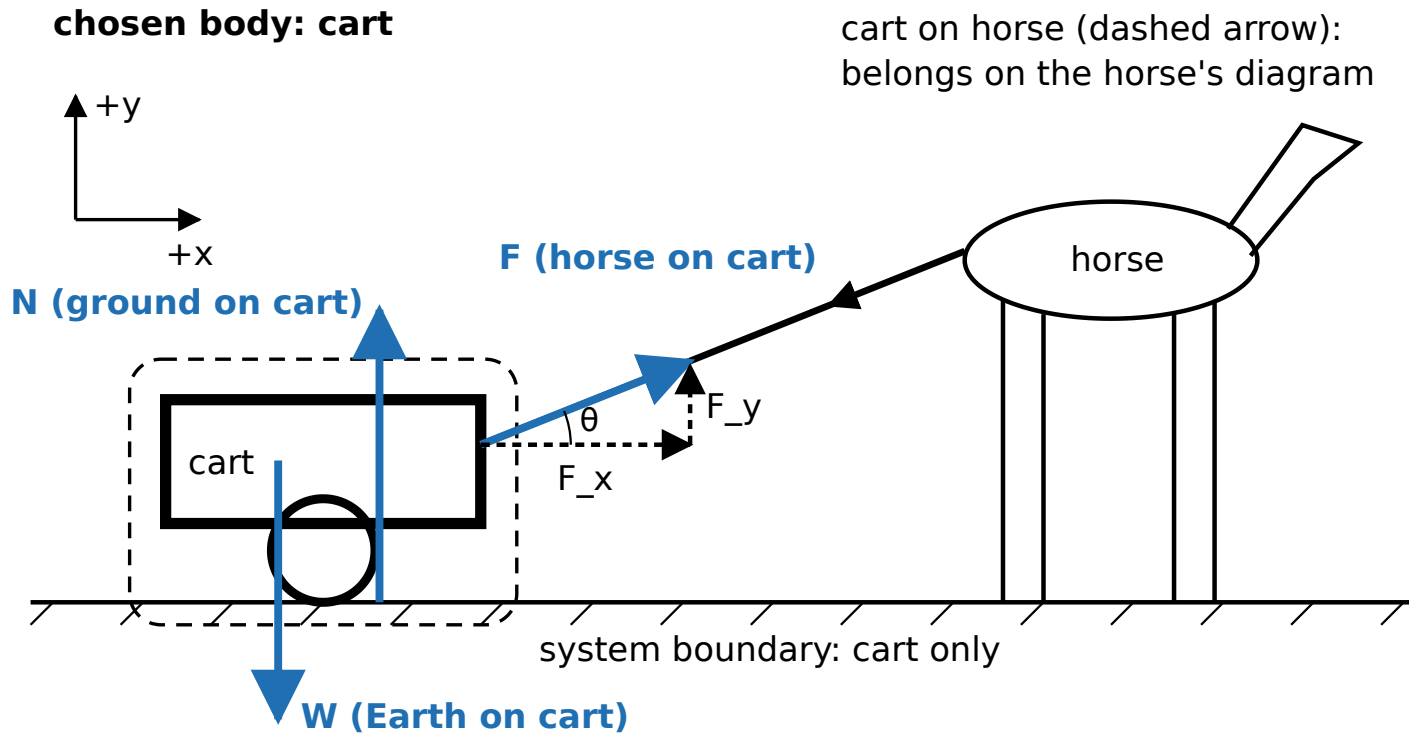
What a complete answer does

Each diagram contains only the member of the contact pair acting on its chosen body.; The partner arrows are placed on different body diagrams.

- 'Cart: horse pushes cart forward. Horse: cart pushes horse backward.' The two arrows are on different diagrams.

The step you rebuilt

A free-body diagram contains forces acting on one chosen body only; the equal-and-opposite partner of an interaction acts on the other body and belongs on that other body's diagram.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Boundary test

Two hands push the same box equally in opposite directions. Can those two forces both appear on the box's free-body diagram?

Boundary answer

Yes. Both hands act on the same chosen box, so both forces belong there and may cancel in the net force.

The rule is body ownership, not a ban on opposite arrows appearing together.

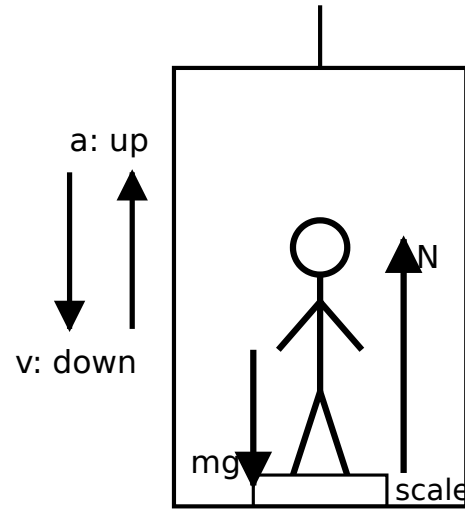
An accelerating observer needs a declared pseudo-force convention

Predict

A bus starts forward and a loose object seems to shift backward to a passenger. Before adding any new force, did a new physical agent suddenly begin pulling the object backward?

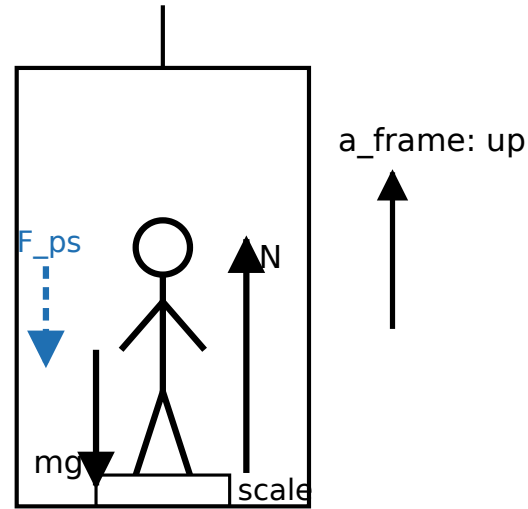
Ground observer (inertial)

$a_{\text{frame}} = 0$



Lift observer (accelerating frame)

person: $v = 0$, $a = 0$ relative to lift



chosen body: the person (same event in both columns)

Real (both): N (scale on person), up;
 mg (Earth on person), down.

$$N - mg = m a_y$$

contact: $N > 0$ (on the scale)

Dashed: $F_{ps} = -m a_{\text{frame}}$, down;
a frame term, no agent, no partner.

$$N - mg - m a_{\text{frame}} = 0$$

contact: $N > 0$ (on the scale)

Same observable in both frames: scale reads $N = m(g + a_y) > mg$.

Set by the upward acceleration, not by the downward velocity.

Show one physical event in an inertial and an accelerating observer description while keeping physical interactions fixed, making the pseudo-force visibly frame-dependent, and preserving the same observable scale reading or string direction.

Attempt

Two labelled observer descriptions of the same object: roadside and accelerating bus, with a statement of where the pseudo-force convention appears.

Reconstruct

1. Did the set of physical objects touching or attracting the loose object change when the bus started?
2. Which observer is accelerating during the bus start?
3. What does the passenger see the loose object do relative to the bus?
4. Where should the pseudo-force be added so the two descriptions remain consistent about the same physical situation?

Diagnose

- A loose mass appears to accelerate backward inside a bus that starts forward. Can the bus observer use only the same physical-force inventory as the roadside observer?

Repair

- State the observer first. Use physical interactions alone for a non-accelerating observer; for an accelerating observer include the corresponding pseudo-force consistently.

What a complete answer does

The roadside description keeps only physical interactions.; The bus description adds the pseudo-force specifically because that observer accelerates.

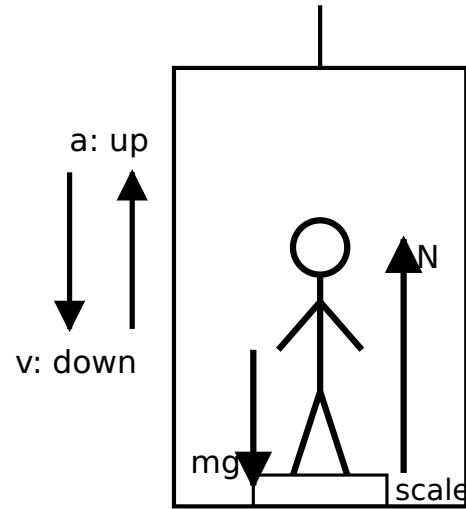
- 'Roadside: physical forces only. Accelerating bus: same physical forces plus a backward pseudo-force for the bus description.'

The step you rebuilt

A force-based account depends on the observer: a non-accelerating observer uses the physical interaction forces, while an accelerating observer must add the corresponding pseudo-force convention to describe the same observed motion consistently.

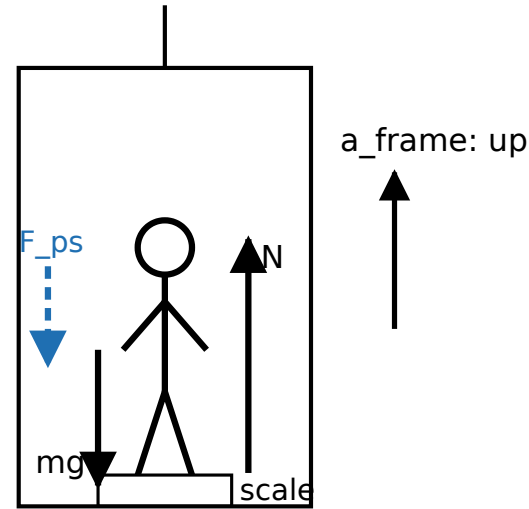
Ground observer (inertial)

$a_{\text{frame}} = 0$



Lift observer (accelerating frame)

person: $v = 0$, $a = 0$ relative to lift



chosen body: the person (same event in both columns)

Real (both): N (scale on person), up;
 mg (Earth on person), down.

$$N - mg = m a_y$$

contact: $N > 0$ (on the scale)

Dashed: $F_{ps} = -m a_{\text{frame}}$, down;
a frame term, no agent, no partner.

$$N - mg - m a_{\text{frame}} = 0$$

contact: $N > 0$ (on the scale)

Same observable in both frames: scale reads $N = m(g + a_y) > mg$.

Set by the upward acceleration, not by the downward velocity.

Stage 1 of 5

Show one physical event in an inertial and an accelerating observer description while keeping physical interactions fixed, making the pseudo-force visibly frame-dependent, and preserving the same observable scale reading or string direction.

Boundary test

The bus now travels at constant velocity. Does changing from the roadside observer to the bus observer by itself require the added pseudo-force?

Boundary answer

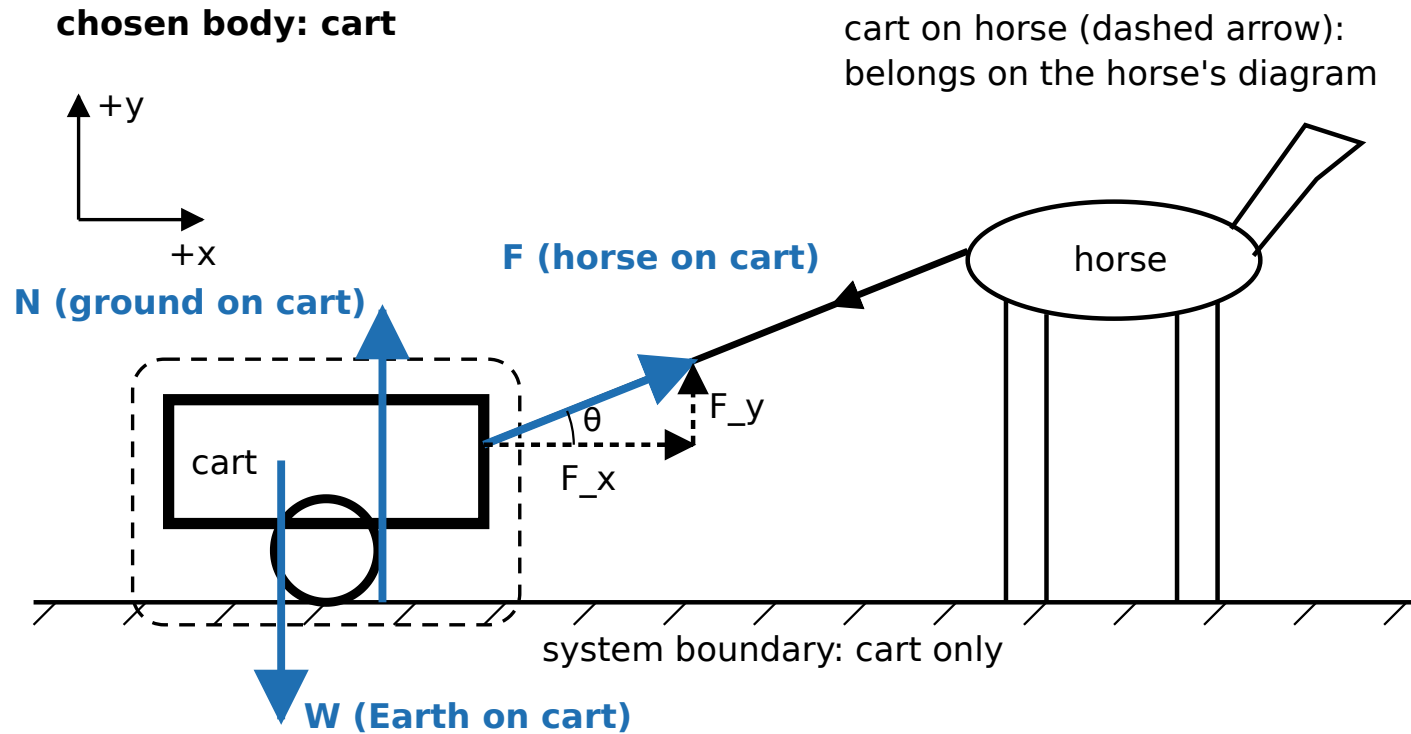
No. With no observer acceleration in this comparison, the extra pseudo-force convention is not required merely because the observers move relative to each other.

The trigger is observer acceleration, not observer motion in general.

Net force, not any one force, sets acceleration

Predict

A body moves right while its net external force points left. Which way is its acceleration?



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Attempt

A one-body signed force sum and a signed acceleration.

Reconstruct

1. Which body are we solving for?
2. What is the signed sum of forces on that body?
3. What positive mass belongs to that same body?
4. Does the zero-net-force boundary give zero acceleration?

Diagnose

- Two 10 N forces act opposite each other on a body. Does either force make the body accelerate in its direction?
- Can a block remain in contact with a fixed plane while speeding up down the plane?

Repair

- Add all signed external forces first; acceleration follows the net force, not an isolated arrow.

- Yes. Maintained contact sets only $a_{\text{perp}}=0$. The parallel component a_{parallel} is determined independently by the net force along the plane.

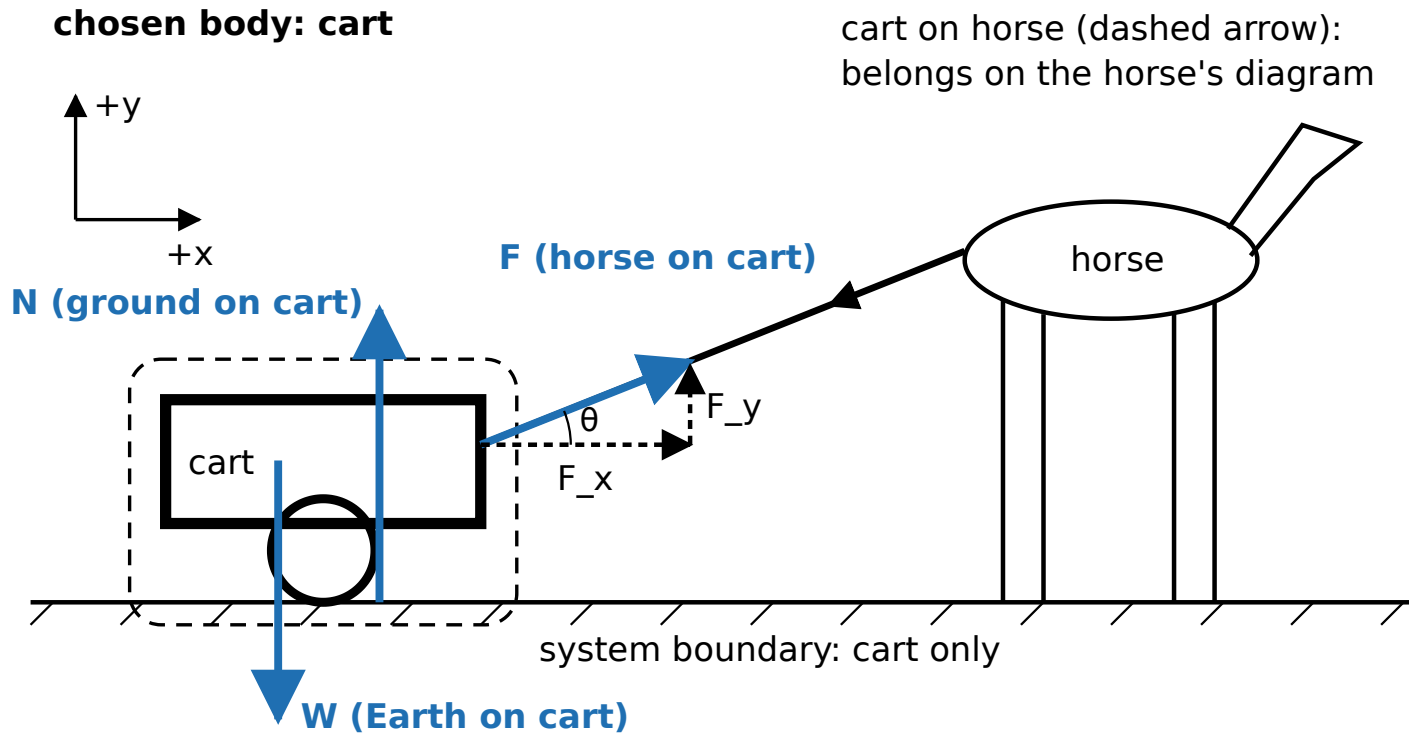
What a complete answer does

Sums forces before using $F = ma$.; Keeps the acceleration sign consistent with the chosen axis.

- +12 N on 4 kg $\rightarrow$ +3 m/s².

The step you rebuilt

For a chosen body of positive mass in the Grade 9 inertial-frame model, acceleration points with the signed net external force and satisfies $F_{\text{net}} = m a$.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Boundary test

If the net external force is zero on a positive-mass body, what acceleration does $F_{\text{net}} = ma$ give?

Boundary answer

Zero acceleration.

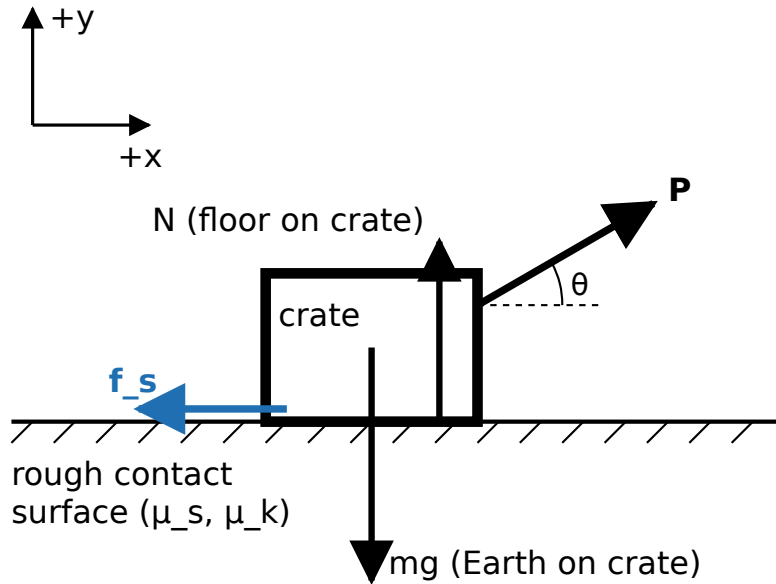
The second-law relation is consistent with the earlier first-law constant-velocity case.

Friction opposes relative sliding at a contact

Predict

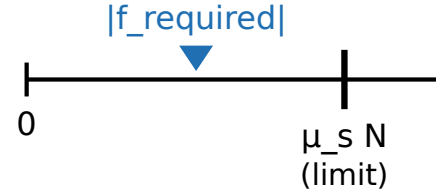
Must friction on a moving object always point opposite the object's motion?

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



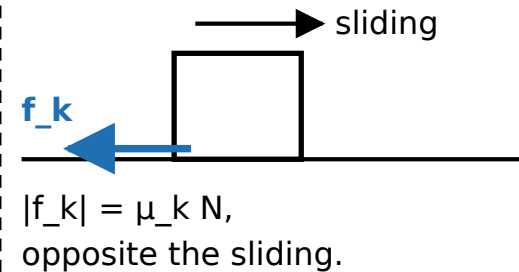
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the kinetic model only after sliding.

Attempt

A contact-level slip-tendency statement and a friction arrow.

Reconstruct

1. Which two surfaces are in contact?
2. Which way would those surfaces slide relative to each other without friction?
3. Which way must friction oppose that relative motion?
4. Does that direction have to oppose the box's ground-frame velocity?

Diagnose

- A moving conveyor belt speeds up a box resting on it. Can the friction on the box point in the same direction as the box's motion?
- A crate and rough platform already move together at constant velocity. If no horizontal force is otherwise required, must the friction force be nonzero merely because the surfaces are rough?

Repair

- Decide the relative sliding tendency at the contact first; friction opposes that tendency.
- No. Ask what relative sliding or slip tendency would occur without friction. If none exists and no tangential force is required, friction can be zero.

What a complete answer does

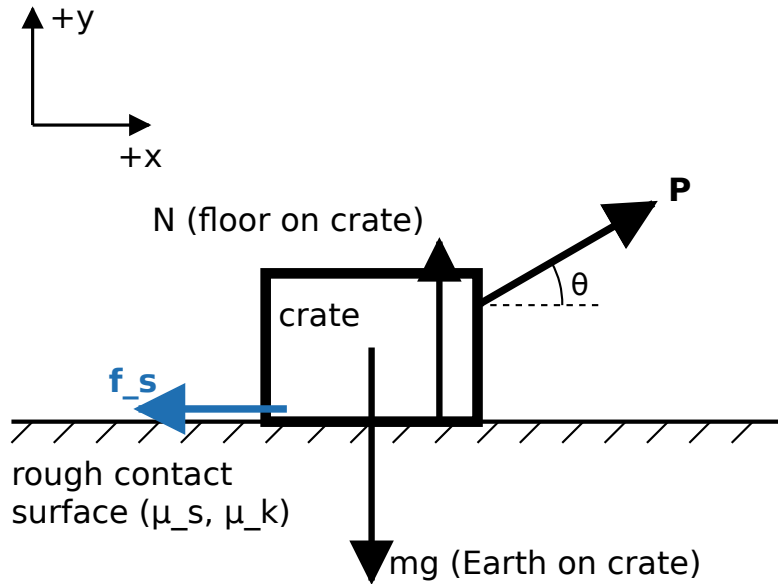
Names the relative slip tendency at the contact.; Points friction opposite that relative slip tendency.

- Belt outruns box -> box slips left relative to belt -> friction on box points right.

The step you rebuilt

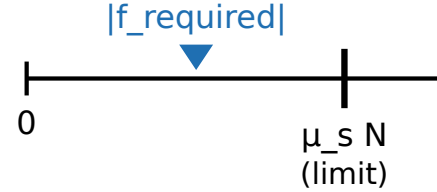
Friction is a contact force along the surface that opposes relative sliding, or the tendency for relative sliding, between the contacting surfaces; it is not defined by the object's overall velocity alone.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



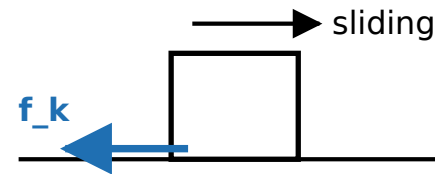
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

Boundary test

If there is no relative sliding and no tendency to slide at a contact, must friction be nonzero?

Boundary answer

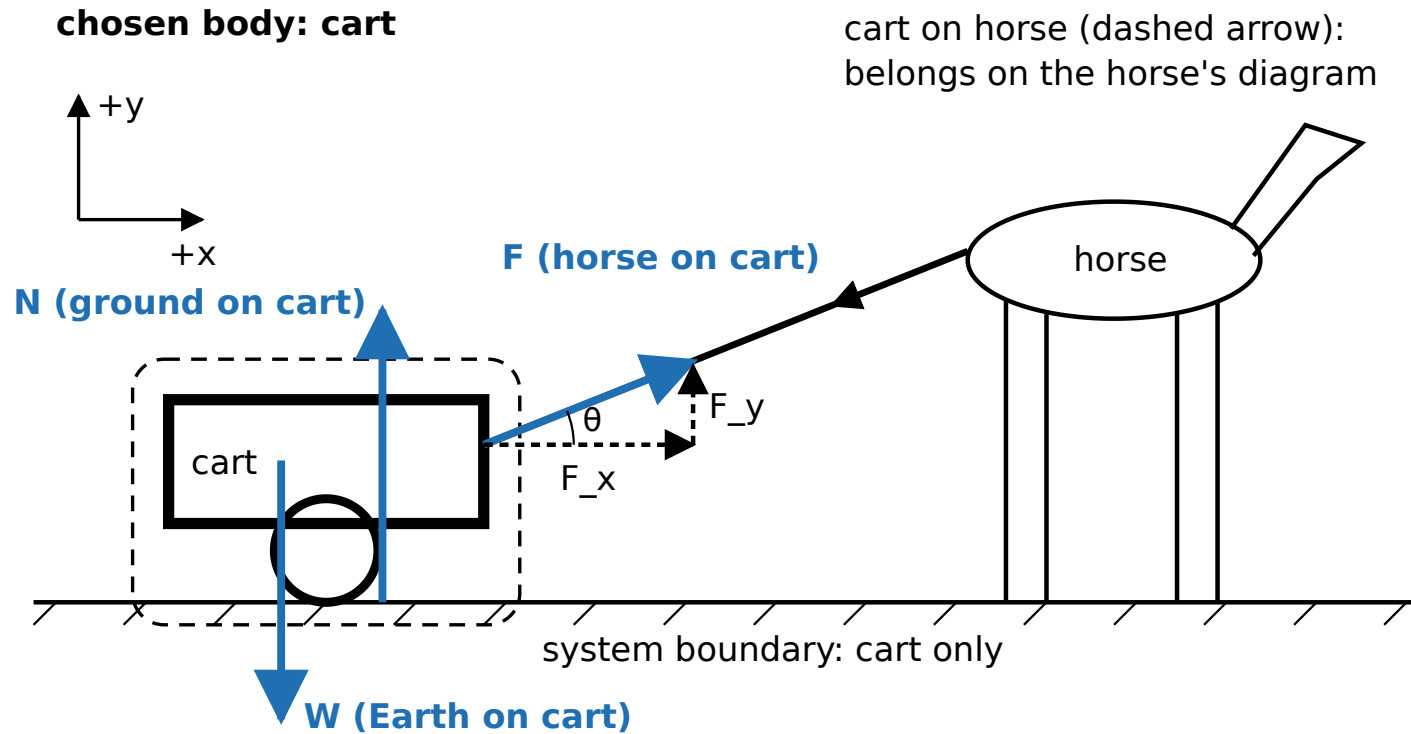
No. Friction can be zero when no tangential contact force is needed.

Friction responds to the contact constraint rather than existing automatically whenever surfaces touch.

Third-law partner forces act on different bodies

Predict

If action and reaction are equal and opposite, why do they not automatically make the net force on one object zero?



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Attempt

Two agent-target force statements and one arrow on each body.

Reconstruct

1. What is the interaction?
2. Which body receives the first force?
3. Which body receives the partner?
4. Can both partners appear in one body's force sum?

Diagnose

- When a person pushes a cart, do the person's force on the cart and the cart's force on the person both appear on the cart's free-body diagram?

Repair

- No. Put each partner on the body it acts on; only then sum forces for one body.

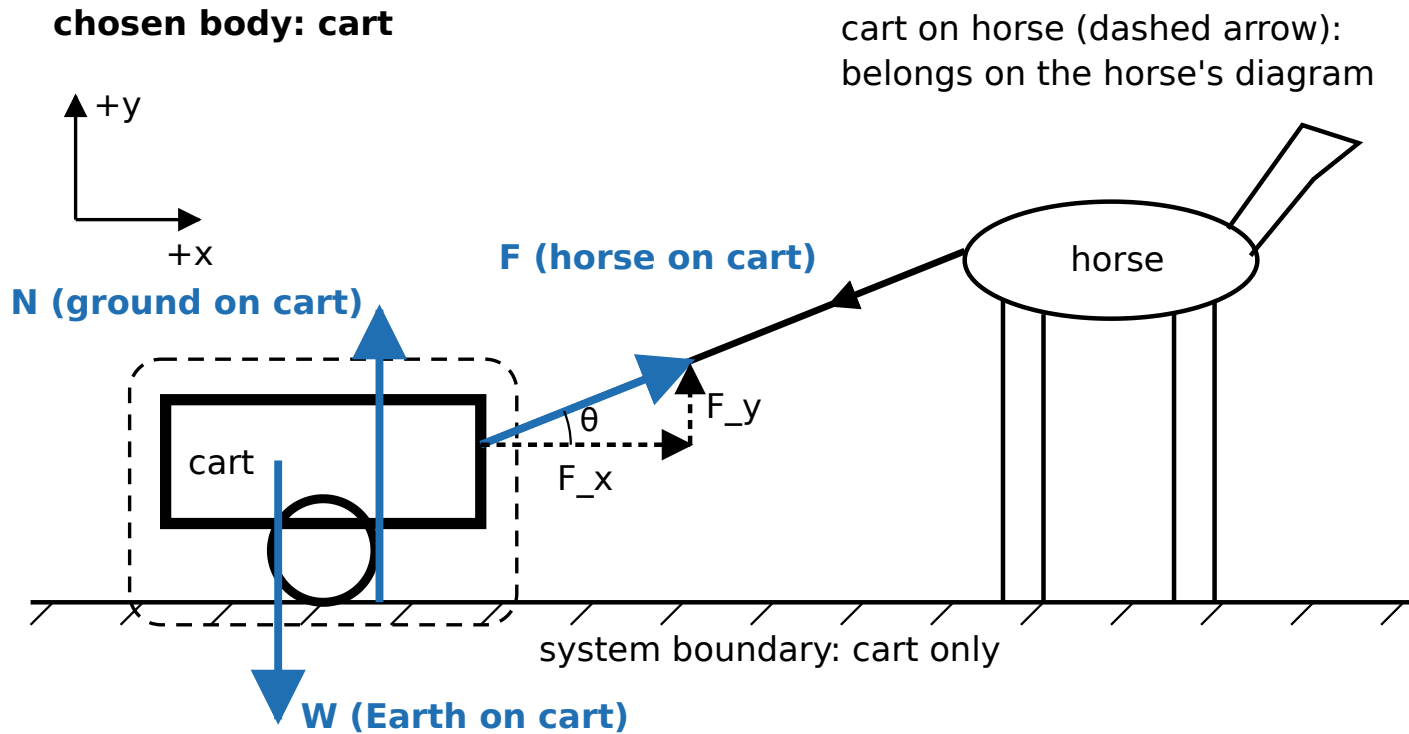
What a complete answer does

Names both partner forces with agent and target.; Places the partners on different body diagrams.

- swimmer on water backward; water on swimmer forward; different bodies.

The step you rebuilt

One interaction produces two partner forces of equal magnitude and opposite direction, but they act on different bodies and therefore do not cancel in a single body's net-force sum.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Boundary test

Can a third-law partner force ever be the force that balances another force on the same body?

Boundary answer

A separate force may happen to balance it, but its own third-law partner acts on the other body and is not the balancing term on the same diagram.

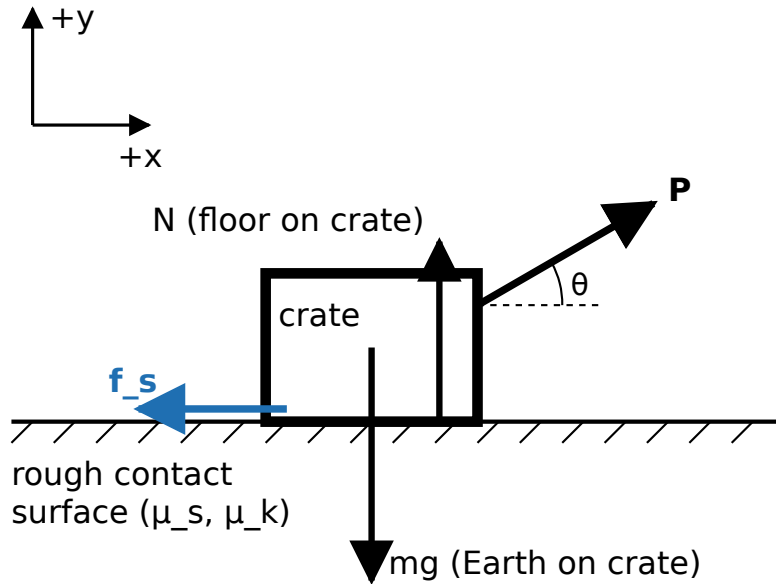
Equal-and-opposite third-law pairing is different from same-body force balance.

Quantitative static and kinetic friction conditions

Predict

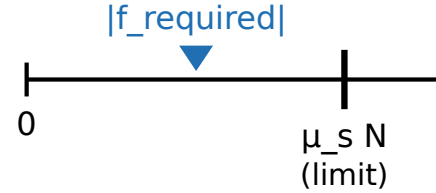
A block remains at rest while the applied tangential force is gradually increased. Before impending slip, must static friction already equal $\mu_s N$?

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



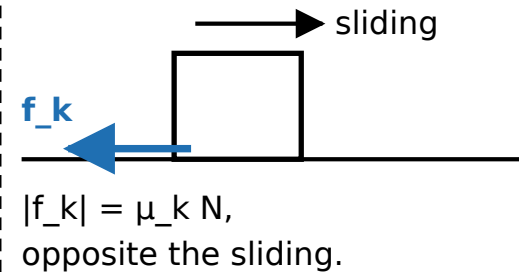
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the kinetic model only after sliding.

Attempt

A contact-state decision that computes the no-slip friction requirement, compares it with the static limit, and selects the correct model.

Reconstruct

1. What normal reaction does the actual perpendicular force equation require?
2. If the contact did not slip, what friction value would the tangential equation require?
3. Is that required magnitude within $\mu_s N$?
4. Which contact state, if any, justifies an equality with a coefficient times N ?

Diagnose

- A small horizontal force is applied to a block that remains at rest. Must static friction already equal $\mu_s N$?
- If an extra force presses a block into a horizontal surface, can the normal reaction still be assumed equal to mg ?
- A problem gives both μ_s and μ_k but does not say whether the contact slips. Can you choose kinetic friction just because μ_k is supplied?

Repair

- No. First solve the force required for no slip. Static friction matches that requirement up to its limit; equality occurs only at impending slip.
- Determine N from the perpendicular free-body equation. $N = mg$ is only a special case.
- No. Determine the contact state first: test whether the no-slip friction demand fits within $\mu_s N$. Use the kinetic model only after sliding is established.

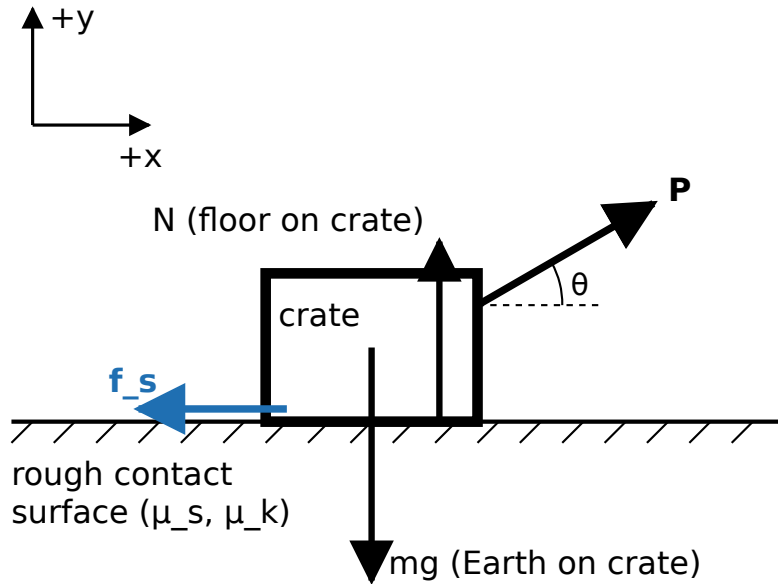
What a complete answer does

Find N from the FBD, solve for f_{required} under no slip, compare $|f_{\text{required}}|$ with $\mu_s N$, and use the kinetic model only after sliding is established.

The step you rebuilt

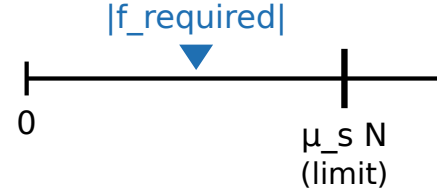
Static friction is a responsive contact force: solve for the friction required to prevent relative slip and compare it with the limiting value $\mu_s N$. Equality is a threshold condition, while the kinetic model is used only after sliding is established.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



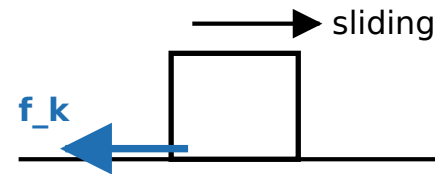
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

Boundary test

If the no-slip equation requires zero friction while $\mu_s N$ is positive, what is the static-friction magnitude?

Boundary answer

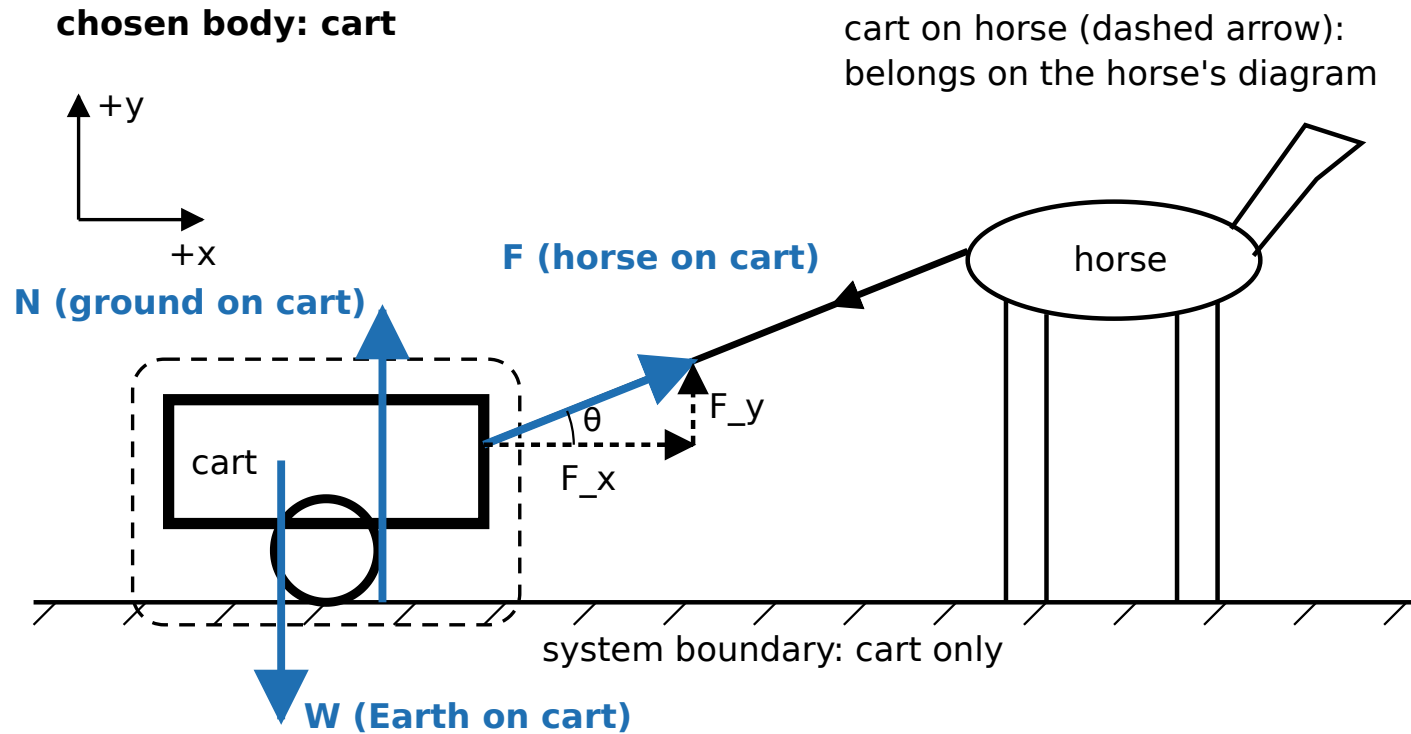
Zero. Static friction need not sit at its maximum.

The static relation is an inequality/response rule, not an unconditional equality.

Connected bodies and a justified common acceleration

Predict

Two blocks move together without separating. Can you assign different accelerations along their common line while their separation stays fixed?



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Attempt

A declared connection constraint plus one signed Newton-II equation per body and an optional combined-system equation.

Reconstruct

1. What physical condition keeps the two bodies' relative separation fixed?
2. With one coordinate direction chosen, what must be true of their constrained acceleration components?
3. Which interaction force belongs in each body's own equation?
4. Only after adding the two body equations, which force pair cancels?

Diagnose

- If two blocks are touching but one loses contact, what keeps their accelerations equal after separation?
- On body A alone, is the force from body B external to A?

- Two connected bodies of different masses share acceleration a . Must their individual net forces be equal?

Repair

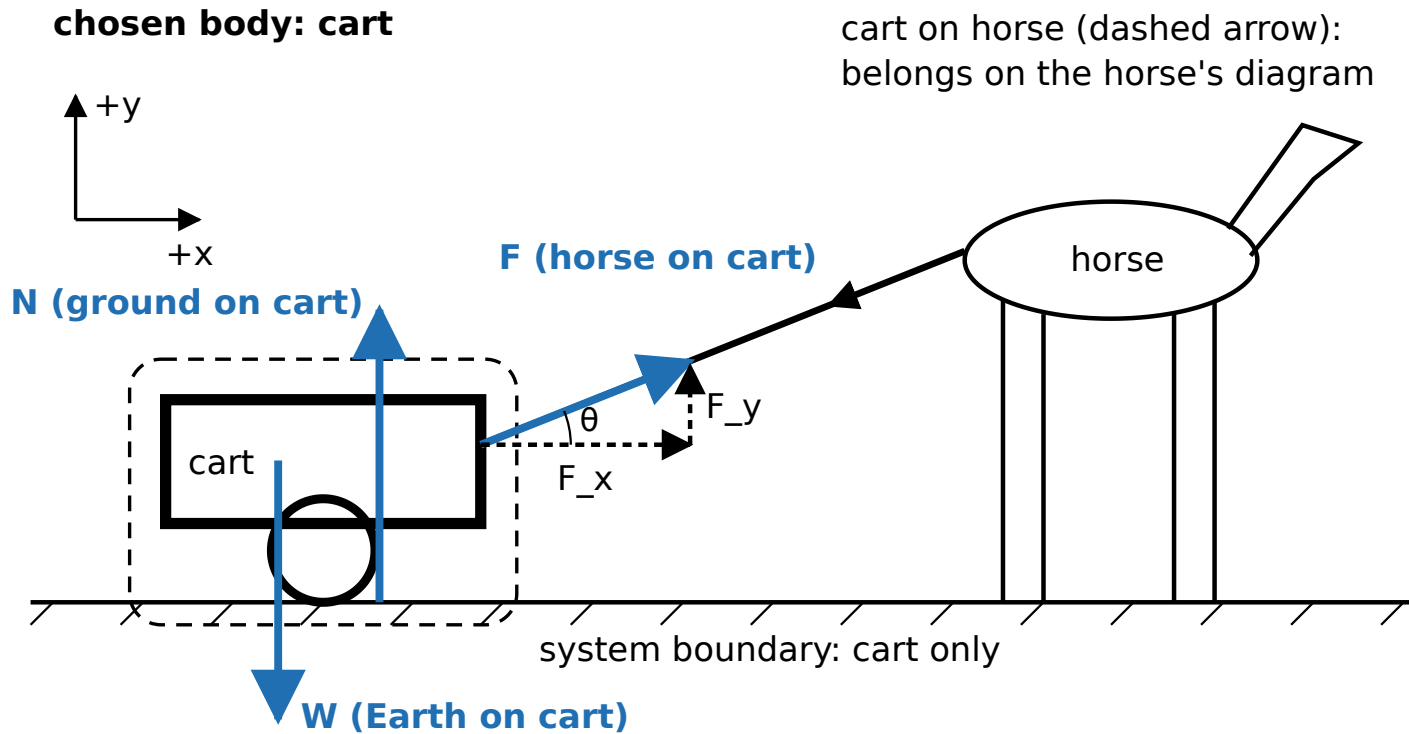
- Name the active constraint first. Equal acceleration is justified only while the relative-separation condition remains enforced.
- Yes. Keep it in A's equation. It cancels only after A and B are deliberately combined into one system equation.
- No. The constraint can make their accelerations equal while Newton II gives different net forces $m_A a$ and $m_B a$ when the masses differ.

What a complete answer does

State the fixed-separation constraint first, use one sign convention, set the constrained acceleration components equal, and cancel internal partner forces only after combining both body equations.

The step you rebuilt

Bodies share an acceleration component only when a physical constraint keeps their relative separation fixed along that coordinate; the shared acceleration can link separate Newton-II equations, while internal partner forces cancel only if the bodies are deliberately treated as one system.



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Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Boundary test

The contact between the bodies is lost. May $a_A = a_B$ still be imposed from the old connection?

Boundary answer

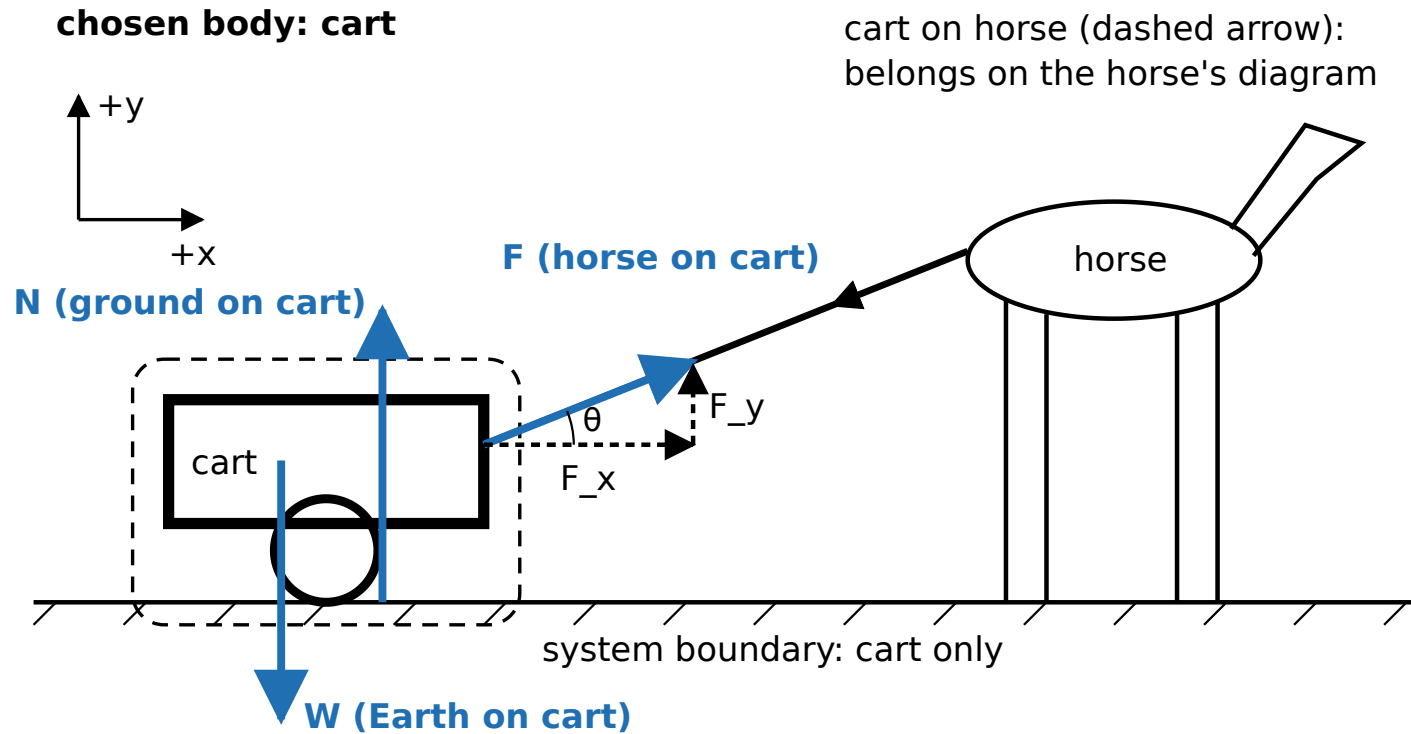
No. Once the constraint no longer keeps their separation fixed, their accelerations must be determined from their new individual dynamics.

Common acceleration is conditional on an active kinematic constraint.

Ideal-string tension belongs to each body-specific FBD

Predict

Two masses hang from the ends of one ideal string over a fixed ideal pulley. May the same symbol T be used for the tension magnitude on both sides?



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Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Attempt

An explicit model declaration plus separate body FBD tension assignments using one common magnitude only when justified.

Reconstruct

1. What assumptions about the string and pulley have actually been declared?
2. Which body does each string force act on?
3. Under those assumptions, can the two tension magnitudes be represented by one T ?

Diagnose

- Do the two end tensions act on the same chosen body?
- What model assumptions justify using one symbol T throughout the rope?

Repair

- No. Put the tension force acting on each body in that body's own FBD. Equal magnitudes across the ideal string do not make them a cancelling pair on one body.

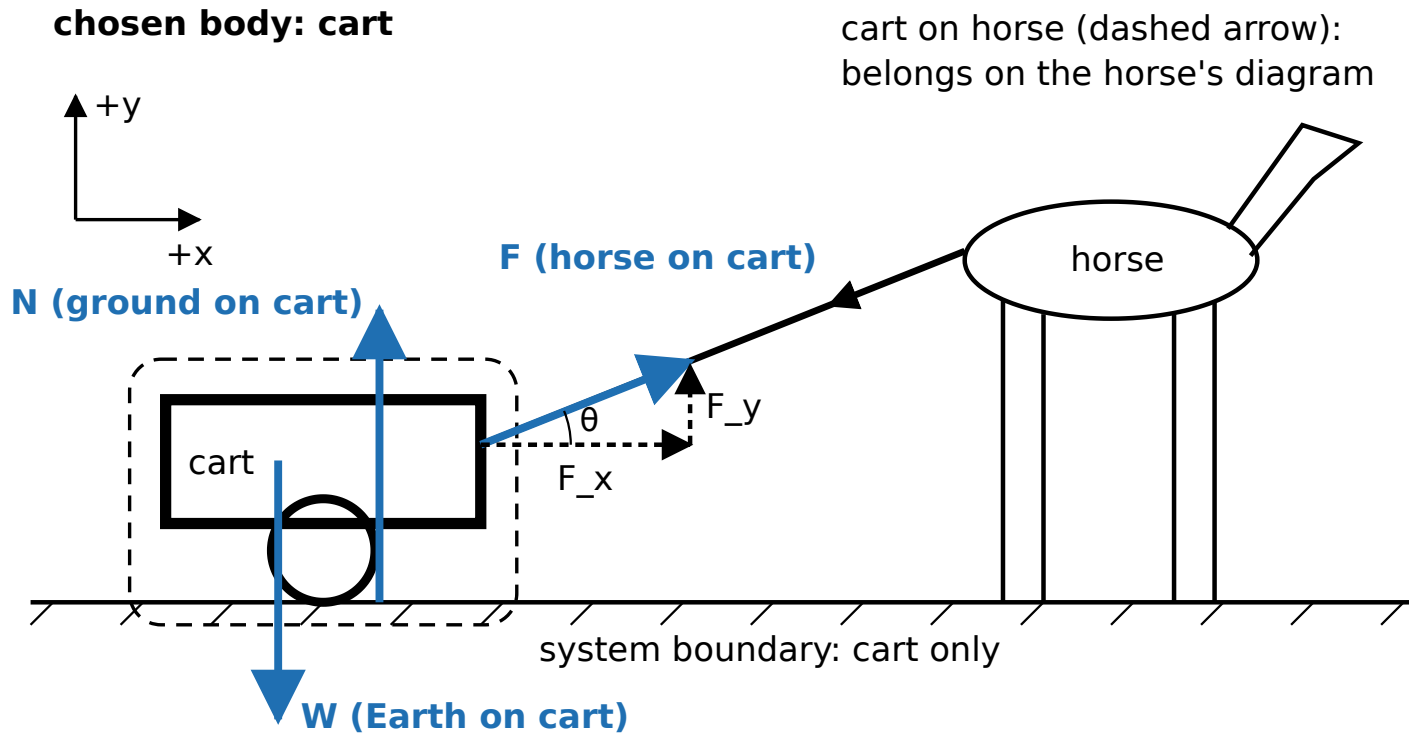
- State the ideal massless string and frictionless/massless redirection assumptions first. Without them, equal tension is not automatic.

What a complete answer does

Declare the ideal string/pulley assumptions, put a tension arrow on each affected body along the local string, and use $T_A = T_B = T$ only under those assumptions.

The step you rebuilt

For one taut massless ideal string, with any redirection treated as frictionless and massless, the tension magnitude can be represented by one T throughout that same string; each body still receives its own tension force along the local string direction.



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Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Boundary test

The pulley axle has significant friction. Does this capability still guarantee equal tension on the two sides?

Boundary answer

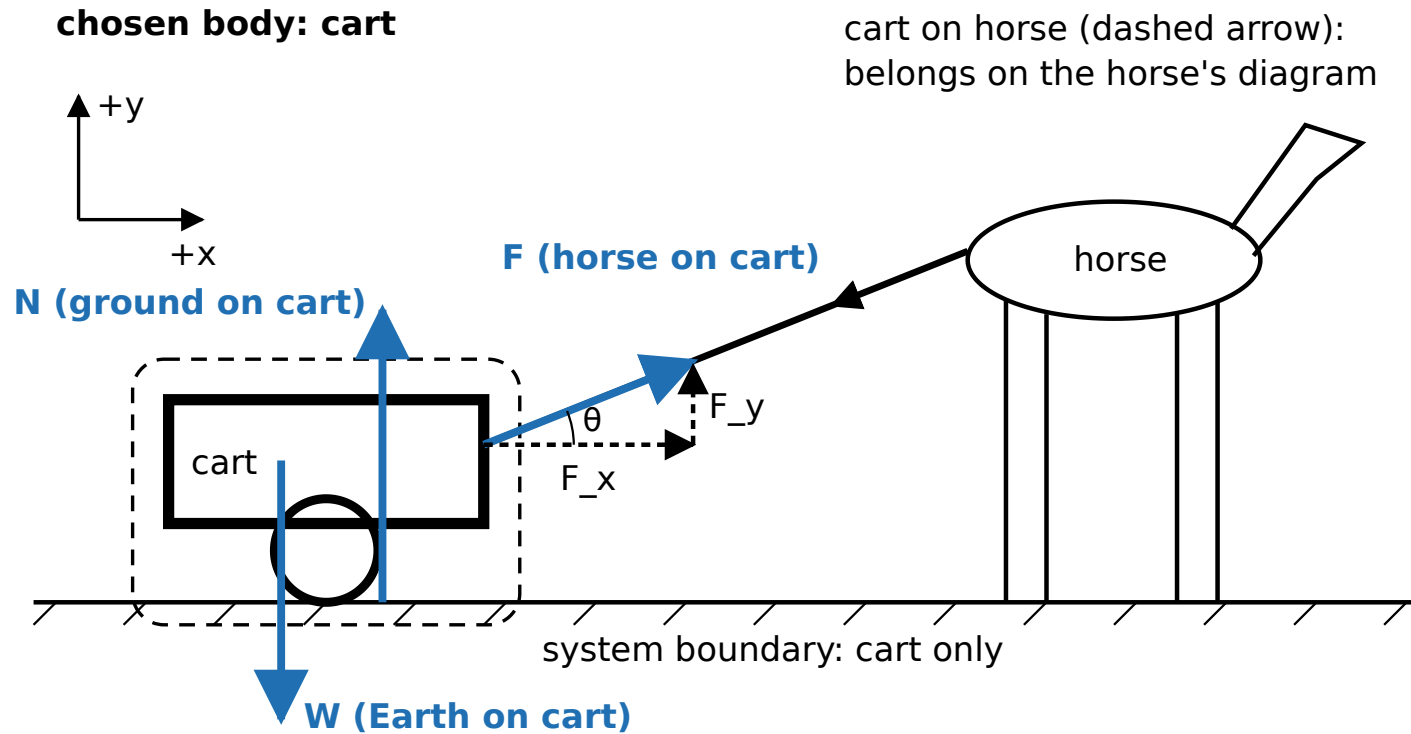
No. The ideal redirection assumption has been broken, so equal tension is no longer guaranteed by this model.

Uniform tension is model-conditional, not universal.

One fixed-pulley string gives one length constraint

Predict

If mass A moves 1 cm farther from a fixed pulley on one side of a taut inextensible string, what must happen to the straight segment on the other side?



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Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Attempt

A coordinate declaration, fixed-length equation, and derived velocity/acceleration relation for one fixed-pulley string.

Reconstruct

1. Which portions of this one string can actually change length while the pulley stays fixed?
2. What equation states that their sum is fixed?
3. What does that equation imply for signed accelerations in the same coordinates?

Diagnose

- What part of the string length changes in this exact geometry, and is any pulley itself moving?
- If both coordinates are positive away from the pulley, can both segment lengths increase while total string length stays fixed?

Repair

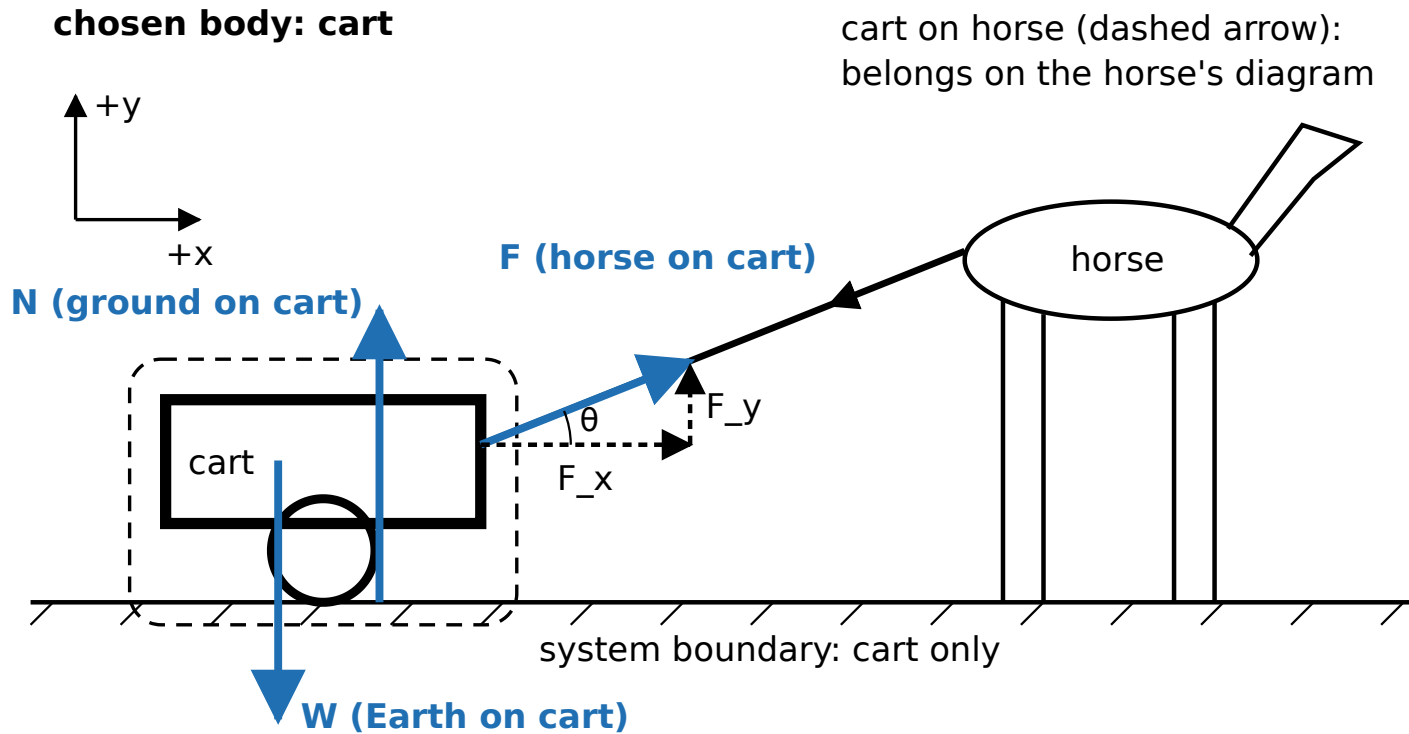
- Write the string-length equation from the actual geometry. The equal-magnitude result here is only for one fixed pulley with two changing straight segments.
- No. Their signed changes are opposite: $a_A + a_B = 0$ in these coordinates.

What a complete answer does

Choose both coordinates away from the pulley, write $y_A + y_B = \text{constant}$, then infer $v_A + v_B = 0$ and $a_A + a_B = 0$.

The step you rebuilt

For one taut inextensible string over one fixed ideal pulley, the two changing straight segment lengths add to a constant; with coordinates measured away from the pulley this gives $y_A + y_B = \text{constant}$ and therefore $a_A + a_B = 0$.



External agents acting on the cart: horse, Earth, ground

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Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Boundary test

A pulley itself is allowed to move. Can you keep using $y_A + y_B = \text{constant}$ without re-deriving the geometry?

Boundary answer

No. A movable pulley changes which string segments vary, so the length equation must be rebuilt.

This capability is intentionally bounded to one fixed pulley.